

Career Summary

- I have completed my **PhD** in the Department of ECE at UCF in **December 2024** with a concentration in ML/AI.
- I have **5+ years** of experience in **designing, developing, and implementing** Deep learning models like **LLMs, Transformers, GNNs, Generative models, Diffusion models, Foundation models, Multimodal Fusion models**, and **RL**, on various applications including **NLP, Bioinformatics, Healthcare, Forecasting, Electronics, CV**, and have familiarity with **version control(GIT), SQLite, MySQL, PostgreSQL, cloud computing(AWS), Parallel/Distributed computing (Multi GPU training), Docker, RAG, Prompt Engineering, MCP, and Data curation and Processing (Pandas, SQL, PowerBI, Kafka, Spark, Hadoop, Tableau)**.
- I have **6 years** of experience in the **Python (NumPy, SciPy, Pandas), java, and C++** programming languages using **TensorFlow, Pytorch, HuggingFace, Scikit-learn, RDKit** frameworks, along with a deep understanding of **statistics** and **linear algebra**, and proficiency in clean, scalable, and documented coding practices.
- I have multiple publications in **top ML journals and conferences** and have experience working in a research environment.
- I also have industry experience during my internship as a Data Scientist at **Bayer** as well as working on joint projects with **Tokyo Electron, ARM, and Synopsys** on real-world ML applications and large scale applied ML projects.

Experience

Bayer Research and Development Services

Missouri, USA

Remote Sensing Data Science Intern

May 2023 - Aug 2023

Project:

- Determining Phenotype of corn plants from UAV (Unmanned Ariel Vehicle) images of corn fields.
- Implemented various CNN and transformer-based vision algorithms and evaluated and compared their performance to identify the best-suited model for the application.

University of Central Florida (UCF)

Florida, USA

Research Assistant

Aug 2019 - Present

Primary Responsibility includes working on various ML related research Projects. Some of them are:

- KITE-DDI**: A Knowledge graph integrated Transformer architecture for accurately predicting Drug-Drug Interaction events from SMILES
- ProtEC**: A Transformer based deep learning model to accurately predict Enzyme Commission numbers from full-scale protein sequences
- ProtGO**: A Transformer based fusion model for accurately predicting Gene Ontology(GO) terms from full scale protein sequences.
- Collaborative Project with Tokyo Electron Ltd. to develop Design Flow for 3D IC design

Selected Research Projects

KITE-DDI

UCF, USA

A Knowledge graph integrated Transformer model for accurately predicting Drug-Drug Interaction Events

- A Knowledge Graph and transformer based fusion model have been proposed which could predict DDI events with SOTA accuracy.
- Shows robust performance with little dependency on sequence length and training set size.
- Paper Link: <https://ieeexplore.ieee.org/document/10909071>
- GitHub Repository: <https://github.com/azwad-tamir/KiteDDI>

ProtEC

UCF, USA

A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers

- A Bert-based Transformer model to predict all four Enzyme Commission numbers from full-scale Protein sequences.
- Pretrained ProtBert modules were selectively finetuned on the Uniprot SwissProt protein database to achieve SOTA accuracy.
- GitHub Repository: https://github.com/azwad-tamir/Prot_EC
- Paper Link: <https://ieeexplore.ieee.org/document/10238761>

ProtGO

UCF, USA

A Transformer Based Fusion model for accurately predicting Gene Ontology (GO) Terms from full-scale protein sequences.

- A Bert-based Transformer model to annotate full scale protein sequences with Gene Ontology (GO) terms using a classification type parallel ML fusion architecture.
- The proposed model is able to predict GO terms with better accuracy compared to other state-of-the-art methods and could understand motifs and protein 2D structures.
- GitHub Repository: <https://github.com/azwad-tamir/ProtGO>
- Paper Link: <https://arxiv.org/abs/2412.05776>

Skin Cancer Detection

UCF, USA

Medical Imaging using ML

- Several Deep Learning classification Model was implemented to detect skin cancer from images of skin lesions.
- Transfer Learning was used on different Vision Transformer (ViT) models to classify skin cancer types on the HAM10000 dataset.
- ViT-based models showed significant improvement over the CNN models.
- GitHub Repository: https://github.com/azwad-tamir/Transfer_ViT

Education

University of Central Florida (UCF)

Orlando, Florida, USA

PhD in Electrical Engineering

Fall 2019 - Fall 2024

- **Research Areas:** Developing various Deep-learning Models in NLP, CV, and GNN for low data Applications in various real-world applications.
- **Dissertation Title:** Architectural and Training regime Modifications to Transformer models for Low Data Applications
- **Current CGPA:** 4.0 out of 4.0
- **Courses:** Machine Learning, Machine Learning Methods for Biomedical Data, Computational Genomics, Data Mining Methodology II, Intro to Neural Networks, Computer Vision, 3D Computer Vision, Design & Analysis of Algorithms, CMOS Analog & Digital Circuit Design, Malware Software Vulnerability, Incident Response Technologies

University of Central Florida (UCF)

Florida, USA

Masters of Science in Electrical Engineering

Aug 2019 - Spring 2023

- **CGPA:** 4.00 out of 4.00
- **Major:** Electrical Engineering

Selected Publications

A. Tamir and J. -S. Yuan, "KITE-DDI: A Knowledge Graph Integrated Transformer Model for Accurately Predicting Drug-Drug Interaction Events From Drug SMILES and Biomedical Knowledge Graph," in IEEE Access, vol. 13, pp. 40028-40043, 2025, doi: 10.1109/ACCESS.2025.3547594.

A. Tamir, M. Salem and J. -S. Yuan, "ProtEC: A Transformer Based Deep Learning System for Accurate Annotation of Enzyme Commission Numbers," in IEEE/ACM Transactions on Computational Biology and Bioinformatics, vol. 20, no. 6, pp. 3691-3702, Nov.-Dec. 2023.

A. Tamir and J.-S. Yuan, "ProtGO: A Transformer based Fusion Model for accurately predicting Gene Ontology (GO) Terms from full scale Protein Sequences." arXiv, 2024. doi: 10.48550/ARXIV.2412.05776. Available: <https://arxiv.org/abs/2412.05776>

A. Tamir, E. Watson, B. Willett, Q. Hasan and J. Yuan, "Crime Prediction and Forecasting using Machine Learning Algorithms," International Journal of Computer Science and information Technologies, Vol. 12(2), 2021, pp. 26-33, ISSN:0975-9646.

A. Tamir, C. Jahan, M. Saif, S. Zaman, M. Islam, A. Khan, S. Fattah, and C. Shahnaz, "Detection of anemia from image of the anterior conjunctiva of the eye by image processing and thresholding," 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), Dhaka, 2017, pp. 697-701. doi: 10.1109/R10-HTC.2017.8289053.

Skills

Programming	Python (NumPy, ProtoBuf, Pandas, etc.), R, C/C++, Bash, Java, Matlab/Simulink, Verilog, SQL, PowerBI, Kafka, Spark, Hadoop
Frameworks	HuggingFace, PyTorch, TensorFlow, Scikit-learn, Keras, AWS
Software	Agile, Synopsys Design & ICC2 Compiler, Lumerical FDTD solutions, Cadence Virtuoso, SPICE, Proteus
Hardware	Embedded Systems, Robotics, Microcontroller (AVR), Arduino, Raspberry Pi, FPGA

Certifications

Stanford University	Machine Learning, course conducted by Coursera
DeepLearning.AI	Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
DeepLearning.AI	Neural Networks and Deep Learning
DeepLearning.AI	Natural Language Processing with Attention Models
DeepLearning.AI	Structuring Machine Learning Projects
Linkedin Learning	SQL Essentials Training
UIUC	VLSI CAD Part II: Layout. Course conducted by Coursera